

1

Temporal network

continuous-time
event stream

$[(u, v, t_1), (u, w, t_2), \dots]$



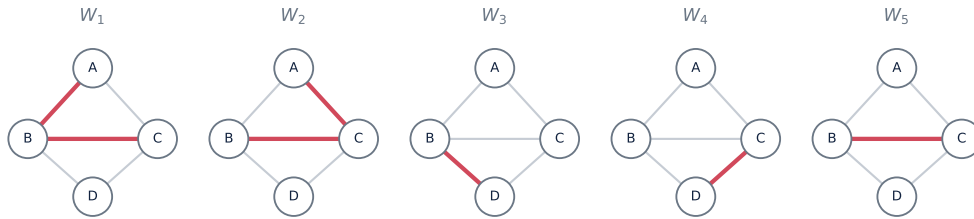
The full pair list is
not the observation.
Access is partial
and can be biased.

2

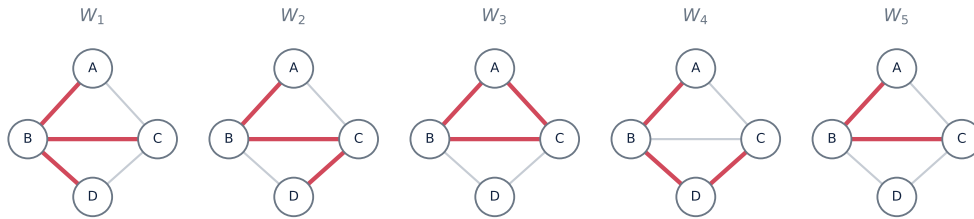
Controlled benchmark

same pairs, different temporal placement $\rightarrow$ different ρ

low persistence ($\rho_2 = 0.20$)



high persistence ($\rho_2 = 0.80$)



K = number of the $W = 5$ windows in which a pair is active
 ρ_k = share of pairs of the FULL network with $K \geq k$, $k = 2 \dots 5$

the profile, not one number: the benchmark scores all
four levels, with $k = 2$ as the headline slice

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Partial access

7 mechanisms in the
benchmark, 2 families

walk-based

- ● **time-agnostic**
no timestamps; leakage control
- ● **time-agnostic + times**
static hops, observed times
- ● **time-respecting**
CTDNE-style temporal order
- ● **recency-biased**
favours recent events
- ● **multi-start**
several entry points

non-walk

- **recent history, $k = 5$**
complete history, 5 nodes
- **recent history, $k = 20$**
complete history, 20 nodes

the frozen LLM suite uses
three of these: two walks
and one non-walk panel

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Compact summary

what the estimator
actually receives

$(n, \text{window mask})$ **histogram**

per observed pair: how often it was
recorded, and in which windows

n, mask	pairs
1, 01	6
1, 02	19
1, 04	23
2, 01	2
3, 01	1

no hand-built feature vector:
the summary is the raw counts,
so the estimator has to do the
bias correction itself

variants ablate the input:
crawl order, timestamps,
recency, node panel

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Estimation and scoring

three families on the
same cases

analytical estimators

explicit bias corrections

supervised baselines

scikit-learn ExtraTrees
and RandomForest

language models

zero-shot and
disclosed-examples

leakage-aware paired evaluation

$$\text{ProfileMAE} = \frac{1}{4} \sum_{k=2}^5 |\hat{\rho}_k - \rho_k|$$

invalid or missing output is
penalized, never repaired

secondary targets:
mean occupancy, one-step
persistence C , link lifetime